Do not write outside the box

0 7 This question is about equilibrium.

0 7 . 1 Describe how a reaction reaches equilibrium.

[2 marks]

Nitrogen dioxide gas reacts to form dinitrogen tetraoxide gas.

The reaction is reversible.

The equation for the reaction is:

$$2\ NO_2(g) \rightleftharpoons N_2O_4(g)$$

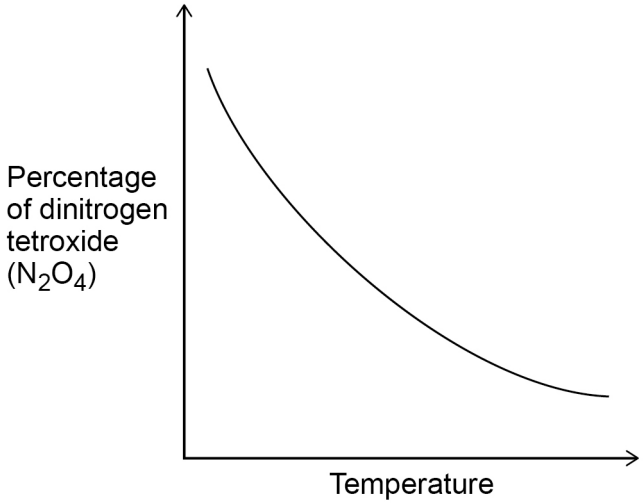
0 7 . 2 Explain the effect on the equilibrium position of increasing the pressure.

[2 marks]

Do not write outside the box

0 7 . 3 **Figure 7** shows the change in the percentage of dinitrogen tetroxide (N_2O_4) in the equilibrium mixture as the temperature of the equilibrium mixture is changed.

Figure 7



Explain the effect on the equilibrium position of increasing the temperature.

Use **Figure 7**.

[3 marks]

_____

_____

_____

_____

_____

_____

7

END OF QUESTIONS